

Reinforcement Learning for Optimal Trade Execution

What an execution agent learns once the evaluation stops flattering it

This replaces an earlier version. The first write-up and a LinkedIn post reported that the agent cut execution cost by about 27% against TWAP and VWAP and 30% against Almgren-Chriss. Those numbers were wrong. They came from a cost metric that charged adverse price moves but never credited favourable ones, and from a market model that charged permanent impact per small trade instead of per order. This document explains how those and several smaller problems were found, and what remains once all of them are fixed.

The problem

An institution that wants to buy 200,000 shares cannot buy them all at once without pushing the price up against itself. Spreading the order across the day avoids that but exposes it to the price drifting away while it waits. Trade fast and you pay *market impact*; trade slowly and you carry *price risk*. Choosing between them is the optimal execution problem.

Cost is measured as **implementation shortfall**: the difference between the average price achieved and the price when the decision was made, in basis points. Because faster trading costs more on average but is less risky, a strategy is really a point on a **cost-risk trade-off**, and the fair way to compare two strategies is at the same level of risk.

What the agent competes against

Each benchmark has one setting that moves it along the cost-risk trade-off. Every setting was swept, and the agent is compared with the best one at its own level of risk.

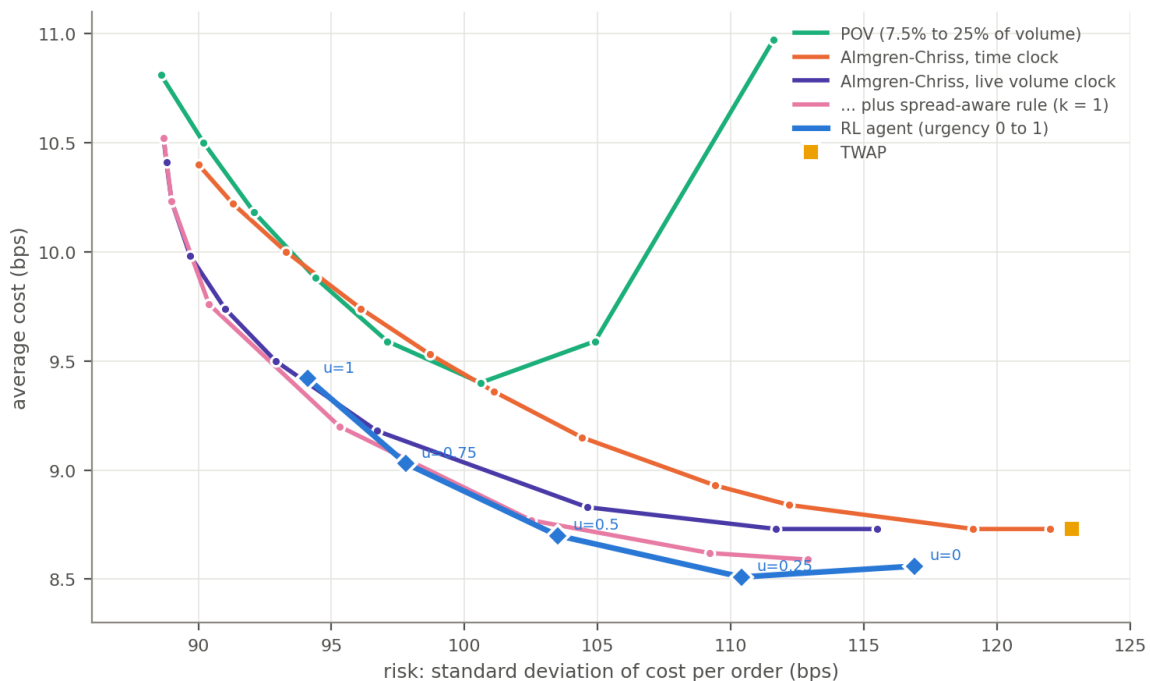
Strategy	What it does	Its dial
TWAP / VWAP	Equal slices per minute / follow the average daily volume curve.	none
POV	Trade a fixed share of whatever volume actually trades.	participation rate
Almgren-Chriss (time clock)	The textbook optimal schedule: front-load to cut price risk, by an amount set by risk aversion.	risk aversion
Almgren-Chriss (volume clock)	The same schedule laid out on the day's volume instead of the clock; the live version scales each minute by the volume actually trading and re-paces on quiet days.	risk aversion
... with a spread rule	Live volume-clock AC that trades less when the quoted spread is wide: $\text{quantity} \times (\text{typical} / \text{current spread})^k$.	risk aversion, k

Result

13 US large-cap stocks, 488 trading days (August 2024 – July 2026), orders of 5% of each stock's daily volume. The data is split into **four walk-forward periods**; in each, **three agents** are trained independently on earlier days and tested only on the following period. Every strategy trades exactly the same episodes. The agent uses only information a real trader has, and runs with a standard schedule guardrail (described below).

Urgency	vs best classical	vs best schedule (no spread rule)	vs Almgren-Chriss (time)	vs POV
0 (patient)	-0.04 (tie)	-0.18	-0.22	-3.78
0.25	-0.11	-0.25	-0.41	-2.30
0.5	-0.08	-0.21	-0.52	-1.01
0.75	-0.01 (tie)	-0.11	-0.56	-0.57
1 (urgent)	+0.07 (tie)	+0.01 (tie)	-0.50	-0.53

Basis points; negative means the agent is cheaper at the same risk. Ties are where the 95% interval, from resampling whole trading days, includes zero. Against plain Almgren-Chriss and POV the agent is cheaper in 10–12 of 12 runs at every urgency.



Each line is one strategy traced across its dial. Down and to the left is better. The agent's urgency dial (blue) sits on top of the best classical curve, live volume-clock Almgren-Chriss with a spread rule (pink), and below every other curve.

In words: trained from scratch, without being told how the market works, the agent learns a cost-risk dial that matches the strongest classical strategy I could build and beats every schedule that ignores the quoted spread, by 0.1 to 0.25 basis points. Those are small numbers: average cost is about 9 bps, and a single order's cost varies by about 100 bps. That is normal for execution, and it is why the evaluation needed paired episodes, four test periods and resampled intervals.

The honest headline: the agent does not beat the textbook. It rediscovers something very close to the best textbook strategy on its own, using only public information. Getting there required finding out why every earlier, more exciting result was wrong.

Against each classical strategy

The headline compares the agent with whichever classical strategy is cheapest at its risk level. Broken down by strategy family, the pattern is clear: the further a strategy is from adapting to live conditions, the larger the agent's lead.

Urgency	AC time clock	AC exp. volume	AC live volume	AC live + spread k=1	AC live + spread k=2	POV
0	-0.22	-0.30	-0.18	-0.04	-0.10	-3.78
0.25	-0.41	-0.43	-0.25	-0.11	-0.17	-2.30
0.5	-0.52	-0.45	-0.21	-0.08	-0.14	-1.01
0.75	-0.56	-0.40	-0.11	-0.01	-0.09	-0.57
1	-0.50	-0.30	+0.01	+0.07	-0.00	-0.53

Agent minus benchmark at equal risk, bps, averaged over 12 runs. The spread-aware rule with $k = 0.5$ sits between $k = 1$ and the live-volume column.

Where the cost comes from

Signed implementation shortfall splits exactly into the market's own drift, the order's permanent impact, execution cost (spread and temporary impact) and fees. Permanent impact is now the same 5.00 bps for every strategy by construction, so all differences come from execution.

Strategy	Permanent	Execution	Fees	Total
TWAP	5.00	3.30	0.43	8.73
VWAP	5.00	3.40	0.43	8.83
Almgren-Chriss (preset)	5.00	3.72	0.43	9.15
POV 10%	5.00	4.16	0.43	9.59
POV 20%	5.00	4.75	0.43	10.18
RL agent (urgency 0.5)	5.00	3.40	0.43	8.84

Mean over all folds and both sides, bps. The agent pays about the same execution cost as TWAP while carrying much less risk (104 vs 123 bps standard deviation), which is what puts it on the frontier rather than above it.

How 27% became a tie

Each row was found by asking why a result looked too good, measuring the cause, and fixing it. Each fix is a few lines of code; finding them was most of the work.

Stage	Apparent edge	What it actually was
Published	~27% vs TWAP, ~30% vs AC	Cost metric ignored favourable price moves; AC was implemented as TWAP
Cost metric fixed	4 bps vs TWAP	Permanent impact charged per small trade, not per order
Impact model fixed	0.9 bps worse than AC	Reward was not the objective; actions clipped at zero
Objective fixed	2.5 bps better than AC	Resting (passive) orders modelled too generously
Aggressive orders only	0.14 bps from the known optimum	Genuine: near-optimal on a problem with a known answer
Impact on live volume	0.3 bps better than best classical	The agent could see hidden simulator state
Public information only	0.3 bps (one test period)	Held in one period, not across four
Four test periods	Tie; worse when patient	One agent in twelve fell behind schedule
+ schedule guardrail	Tie with best; beats schedules	Final result

1. A cost metric that only counted bad luck

Each fill was charged $\max(0, \text{fill price} - \text{arrival price})$. When the price moved against the order the cost counted; when it moved in the order's favour it counted as zero. Random drift averages to zero, but clipped it becomes a cost that grows with time in the market, so slow strategies were penalised regardless of how well they traded. A zero-impact test charged TWAP about 21 bps and a fast schedule about 10 bps from noise alone, the same size as the published edge. Measured properly, TWAP's cost fell from 52.8 to 11.9 bps and the agent's from 39.2 to 8.0. The edge against TWAP shrank from 13.7 to 4.0 bps, and the published loss to POV 20% became a win.

2. Permanent impact charged per trade instead of per order

Permanent impact was measured on real data as strongly concave in trade size (exponent 0.18), and the simulator applied it to every fill. With an exponent that small, a tiny fill moves the price almost as much as a large one, so splitting an order into 390 slices looked expensive: TWAP paid 8.7 bps of permanent impact and the agent's bursts 2.3. The fix makes the price displacement depend on how much of the order has executed (the square-root law for large orders). Total permanent cost is then the same for every schedule that finishes the order, so its level, which public data cannot measure, no longer affects any comparison. The agent's remaining edge disappeared and TWAP became the cheapest strategy on average, as theory predicts.

3. A benchmark that was secretly another benchmark

Almgren-Chriss's risk aversion had been set so small that its schedule was a straight line, identical to TWAP. It was caught because the two agreed to one decimal on all 26 ticker-and-side rows. Every benchmark is now swept across its full range.

4. An agent that switched itself off

The training library samples actions from a bell curve and clips them to the allowed range. Every sample below zero became "trade nothing" with no signal about how far below, so the agent learned to use that region as an off switch: idle in 28% of minutes, then catching up in bursts. Because impact grows faster than trade size, pausing and catching up costs about 1.6 times as much as trading evenly. Mapping the action through a smooth curve instead of clipping it cut idle minutes to 0.1% and brought the agent from 0.8 to 0.14 bps of the known optimum.

A problem with a known answer

Before testing whether the agent can beat classical strategies, it was tested on a version of the simulator where theory says Almgren-Chriss is optimal (impact priced against average volume). The table shows the distance above the Almgren-Chriss frontier at equal risk, bps, one test period, three agents. With the right objective and a smooth action the agent gets within 0.14-0.46 bps of the optimum, and it beats every fixed POV rate by 1-8 bps.

Urgency	0	0.25	0.5	0.75	1
Shaped reward (original)	+0.9	+0.8	+0.9	+0.9	+0.9
Almgren-Chriss reward	+0.93	+0.82	+0.77	+0.81	+0.86
... plus smooth action	+0.46	+0.25	+0.14	+0.14	+0.16

5. Seeing what no trader can see

The agent's inputs included the simulator's hidden volatility regime and news flags. Hiding them removed about three quarters of its edge. They were replaced by what a trader really sees: the current quoted spread and recent price volatility. The quote turned out to be the more useful signal, since it is the cost the agent is about to pay.

6. One good period is not a result

With honest inputs, the agent beat even the spread-aware benchmark on one test period, clearly and significantly. Across all four periods that edge vanished, and at the patient end the agent was worse. Almost all of that came from one of twelve agents, which had finished 39% of its order by three quarters of the day against 75% for a straight-line pace, and then paid 16.5 bps to catch up. A rule that forces the agent to catch up if it falls more than 5% of the order behind a straight-line pace removed the failure without affecting the other urgency levels. Trading desks put controls like this around any model.

Two guardrail widths were tried (5% and 10%) and the better is reported. They give similar results, but it is a choice made after seeing the data.

Testing whether the edge was real

Test (fold 3 unless stated)	u=0	0.25	0.5	0.75	1
Agent sees hidden regime and news flags	-0.12	-0.28*	-0.28*	-0.21*	-0.16*
Same agent, flags hidden	-0.04	-0.07	-0.07	-0.01	+0.04
Same agent, impact ignores spread and volatility	-0.06	-0.15*	-0.14*	-0.11	-0.08
Retrained on public information only	-0.17*	-0.34*	-0.31*	-0.23*	-0.14*
... vs best incl. spread-aware rule	-0.06	-0.23*	-0.20*	-0.14*	-0.09
All 4 periods, no guardrail	+1.71*	+0.47*	+0.06	+0.01	+0.08
All 4 periods, 10% guardrail	+0.13*	-0.05	-0.06	-0.01	+0.07
All 4 periods, 5% guardrail	-0.04	-0.11*	-0.08*	-0.01	+0.07

Agent minus the best classical strategy at equal risk, bps. Rows 1-4 compare with the best strategy that ignores the spread; rows 5-8 include the spread-aware rule. * = 95% interval excludes zero. Each row answers one question: does the edge survive without hidden information, without the uncalibrated impact terms, with a fair benchmark, and in other periods? Most of it did not; what is left is a tie.

The patient-order failure, agent by agent

Fold	Seed 42	Seed 43	Seed 44
0	+0.37 / +0.03	+0.51 / +0.24	+0.38 / -0.09
1	+13.18 / +0.83	+1.28 / +0.15	+2.73 / +0.41
2	+1.22 / +0.32	+0.60 / +0.17	+0.44 / +0.10
3	+0.11 / +0.03	-0.07 / -0.14	-0.26 / -0.49

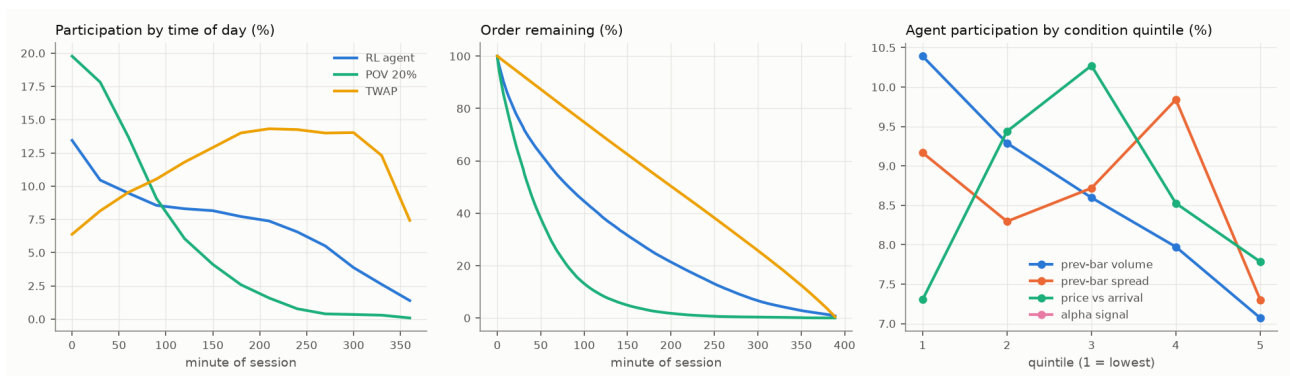
Gap to the best classical strategy at urgency 0, without / with a 10% guardrail, bps. One agent (fold 1, seed 42) fell far behind schedule; the guardrail cuts its excess cost by 94% and barely touches the others.

How the comparison is made

Choice	Why
Signed implementation shortfall	Counts favourable price moves as well as adverse ones; the earlier one-sided metric inflated costs about fivefold.
Paired episodes	Every strategy trades the same stock, day, start time and price path, so differences are not luck of the draw. Checked automatically.
Equal-risk comparison	Each benchmark is swept across its dial; the agent is compared with the lower envelope of each family at the agent's own risk, so a badly tuned setting cannot flatter it.
Walk-forward, 4 x 3	Four test periods, each after its own training window, three independently trained agents per period.
Day-clustered bootstrap	Orders on the same stock and day share a price path, so intervals resample whole days, and rebuild every frontier on each draw.
Model selection	Best checkpoint chosen on the end of the training window, never on the test period.

How the agent works

Part	Design
Simulator	Limit order book driven by real one-minute bars. Temporary impact is measured against the volume trading in that minute; permanent impact follows the square-root law on the cumulative order.
Action	A share of the current minute's volume, passed through a smooth curve so the agent can never accidentally switch itself off.
Inputs	Order and time remaining, price move since arrival, last minute's volume and spread, current quoted spread, recent realised volatility, own recent trading, side, urgency.
Reward	Each minute: execution cost + permanent impact + fees, plus a risk penalty on the inventory still held, weighted by urgency. Summed over the day this is exactly the Almgren-Chriss objective.
Training	PPO (Stable-Baselines3), 10M steps per agent, best checkpoint chosen on the end of the training window, never on test data.
Guardrail	If more than 5% of the order behind a straight-line pace, trade at least enough to close the gap over 10 minutes.



Left: share of each minute's volume traded, by time of day. Middle: order still held. Right: the agent's participation by quintile of recent conditions. The agent front-loads, finishing about 55% of the order by the quarter mark against 25% for TWAP, then tapers, and trades a smaller share of busy minutes than quiet ones.

Grounding the simulator in real data

Two years of tick data (about 47 million trades with the prevailing quotes) were used to calibrate the simulator rather than assume it.

Parameter	Measured	Evidence
Temporary impact exponent	0.65	11.75M classified trades
Permanent impact shape (per trade)	0.18	312M trade-horizon observations
Average spread	2.13 bps	2 years, two independent methods
Activity clustering decay	0.013	6,409 ticker-days
Calm / volatile volatility	0.0004 / 0.0011	2-state mixture
Intraday volume profile, daily volume	measured	per minute, per stock

Measuring impact requires knowing which side started each trade. The standard Lee-Ready classifier was scored against exchange-reported sides: its main rule agreed 99.6% of the time, but its fallback rule scored 51.2% (interval 49.9–52.5%), no better than guessing, so those trades were excluded. Public data only contains small trades (31 of 10.5 million exceeded 1% of daily volume), so the size of impact for large orders cannot be measured; only its shape can.

Limitations

The simulator is synthetic

It is calibrated from real data, but the agent has never traded against replayed order flow, let alone real money. Results are comparisons within the simulator, not predictions of real costs, and training and testing share it.

Still open

Issue	Why it matters
Simulator calibrated on all data	Volume profile, spread and regime parameters were fitted on the full two years, test periods included. The agent never sees test days, but the world it trains in was built with knowledge of them.
Spread variation not calibrated	The edge over spread-unaware schedules comes from simulated spread changes (news windows, volatility). The average spread is measured; how much it varies minute to minute is not yet.
Regime barely moves the price	The simulated volatile regime raises costs but hardly changes the price path, unlike real markets.
Impact size	Not measurable from public data; the temporary impact level is assumed.
Passive orders	Excluded: the fill model was too generous, and the benchmarks cannot rest orders.
Guardrail width	Chosen between two values after seeing results.

Not modelled

Venue routing, queue position, latency, multi-day and portfolio execution, and prices recovering after the order. Hyperparameters were not tuned and the agent was not tested on stocks it never trained on. Six of nineteen stocks were excluded on stated data-quality criteria, so the result describes liquid US large-caps.

What would come next: recalibrate the simulator separately for each test period; measure real minute-level spread variation and match the simulator to it; retrain with more patient-order episodes so the guardrail becomes a backstop rather than a necessity; and test on unseen stocks. Before any real use: replay historical order flow and paper trade.

Code, calibration scripts and the full audit, with every number in this document, are on GitHub. Python, PyTorch and Stable-Baselines3; market data from Databento.